

Watt It Takes: Battery Performance Targets and Pack Sizing for an All-Wheel-Drive Electrified Jeep

Sponsors: [Faculty Sponsor], Department of Electrical Engineering and Computer Science; [EcoCAR Faculty Advisor], UTK EcoCAR Team (liaison to the High Voltage Battery swimlane)

Electrical Engineering (Computer Engineering secondary) • High Voltage Battery • Scored: Battery Performance Targets (Nov 19, 2026), Architecture Selection (Apr 1, 2027)

Project Background and Objective

The team is adding a rear electric drive unit to a 2026 Jeep Cherokee to create an electrified all-wheel-drive off-road vehicle, which carries a consequence nobody can wave away: the stock high-voltage battery was designed to feed one electrified powertrain and is now asked to feed two, while remaining the primary energy source for both through the early competition years. Whether it can — and if not, which limit it hits first — must be answered with numbers before the team commits to an architecture. The analysis runs cleanly from vehicle to cell: acceleration and grade targets become wheel torque, then motor torque and speed, then electrical power at the terminals, then current, then per-cell C-rate, which against the datasheet tells you whether the pack survives a launch on a 32% grade or sags out of the inverter's operating window. Every step is undergraduate electrical engineering with an answer a senior can defend before industry reviewers.

Project Scope

- **Stock system characterization.** System-level schematic covering cells, disconnect unit, BMS, connectors, cooling, and safety features; extract and internally verify the voltage window, energy and amp-hour capacity, discharge ratings, and series/parallel topology rather than trusting the datasheet.
- **Power demand analysis.** Peak and continuous electrical demand for the front powertrain and target rear EDU, combined; the torque-to-power relationship derived explicitly and converted to pack current; short peaks (launch, traction events, obstacle climbing) separated from sustained demand.
- **Sufficiency assessment.** Per-cell C-rate at combined peak against the rated maximum; voltage sag from internal resistance against the inverter's window; sustainable duration; and a plain statement of which constraint binds first — energy, current, thermal, or sag.
- **Pack configuration matrix.** A formula-driven workbook computing every pack parameter from cell inputs, with configurations violating competition electrical constraints flagged automatically.
- **Drive-cycle validation.** Simulate candidate configurations in the virtual vehicle model across representative and off-road cycles, checking energy, power sufficiency, and thermal loading against the analytical predictions.

Technical Considerations

- **Assumptions are the deliverable.** Scoring explicitly penalizes undocumented or unjustified calculations, so every efficiency estimate, rated value, and simplification must be stated, sourced, and defended.
- **Peak and continuous are different problems.** A pack sized only for peak current may lack energy for a long grade; one sized only for energy may sag below the inverter's minimum during launch. Internal resistance is where the interesting physics hides.
- **The workbook must be a live model, not a table.** Changing one parameter should propagate everywhere — a software design problem as much as an electrical one, and what makes the analysis reusable in later years. Cell chemistry trade-offs should be quantitative, not a preference.

Deliverables

- Engineering analysis report in the competition's ten-page format covering characterization, demand, sufficiency, and configuration comparison.
- Formula-driven pack configuration workbook with automatic constraint flagging and documented sources.
- Stock system schematic plus preliminary schematics for candidate team-built topologies, with drive-cycle results supporting the analytical conclusions.
- Recommendation memo on cell type and series-parallel configuration for the Architecture Selection Review, defended in a technical review; report and presentation.

Assumptions and Dependencies

Requires stock battery documentation and sponsored cell specification sheets from Stellantis, preliminary rear EDU power and torque targets from Vehicle Systems Engineering, MATLAB and Simulink with the virtual vehicle model, and Excel for the required workbook format. The work is entirely analytical and simulation-based — no high-voltage hardware handling, so no safety certification gate limits participation. Students need circuit analysis, energy conversion, MATLAB, and spreadsheet modeling; electric machines or power electronics coursework is natural preparation.

Support and Guidance

Mentorship on battery systems, power conversion, and modeling, supported by a short course on battery voltage and capacity definition through MATLAB/Simulink. The program provides high-voltage design training, cell specification data, the USABC Battery Test Manual, and access to Stellantis and Argonne experts. Because the project maps onto scored deliverables with hard deadlines, students get industry critique alongside faculty review.